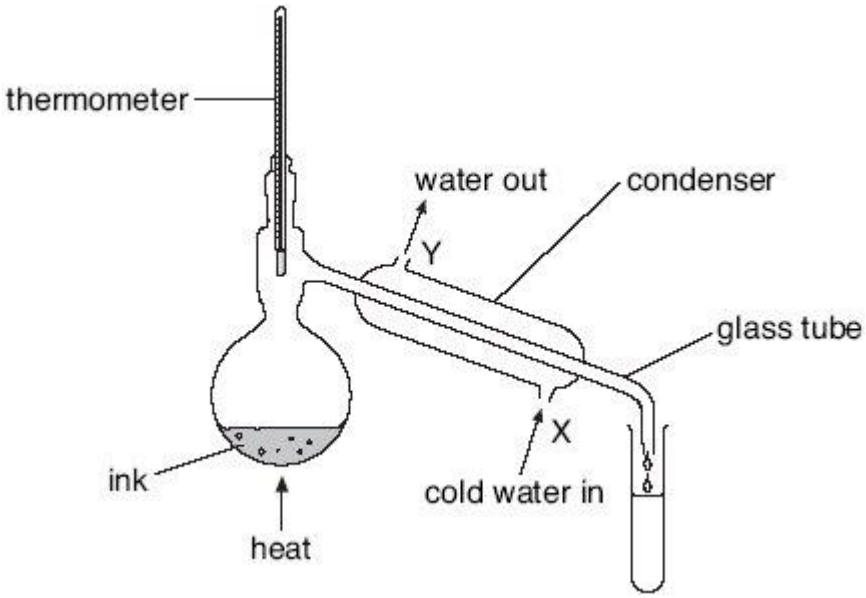
Q1. Rema used the apparatus below to distil 100 cm^3 of water-soluble ink.



apparatus A

not to scale

(a) Which processes occur during distillation?
Tick the correct box.

condensation then evaporation ☐

evaporation then condensation ☐

melting then boiling ☐

melting then evaporation ☐

1 mark

(b) Give the name of the colourless liquid that collects in the test-tube.

.....

1 mark

(c) What would the temperature reading be on the thermometer when the ink has been boiling for two minutes?

.....°C

1 mark

(d) (i) Water at 15°C enters the condenser at X.
Predict the temperature of the water when it leaves the condenser at Y.

.....°C

Explain this change of temperature.

.....

.....

1 mark

(ii) Give **two** ways in which the water vapour changes as it passes down the glass tube in the condenser.

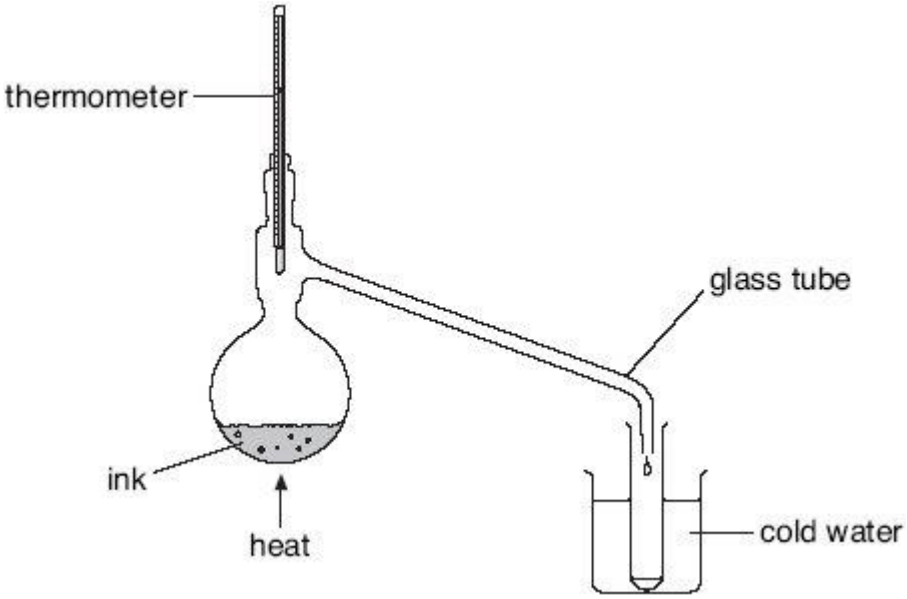
1. .....

1 mark

2. .....

1 mark

(e) Peter used the apparatus below to distil 100 cm^3 of water-soluble ink.



apparatus B

not to scale

Why is the condenser in **apparatus A** better than the glass tube and beaker of water in **apparatus B**?

.....

.....

1 mark

maximum 7 marks

Mark schemes

Q1.

(a) evaporation then condensation ✓

if more than one box is ticked, award no mark

1 (L5)

(b) water

accept 'distillate'

1 (L5)

(c) 100

accept a temperature from 99 to 101

1 (L5)

(d) (i) a temperature above 15 but below 100

any **one** from

- it is heated by the water vapour
  *accept 'vapour **or** steam **or** hot water' for water vapour*
- thermal energy **or** heat is transferred from the water vapour
  accept 'it heats up'
  ***both** the answer and the correct explanation are required for the mark*

1 (L6)

(ii) any **one** from

- it condenses
  accept 'it makes condensation'
- it changes to a liquid
  accept 'it is liquid'
  accept 'it changes state'
  accept 'it turns to water'

1 (L6)

- its temperature falls
  accept 'it cools' 'it changes temperature' is insufficient

1 (L6)

(e) any **one** from

- cold water replaces warm water
  *accept 'it cools the water vapour better **or** more efficiently **or** quickly'*
  accept 'it will work faster'
- the water stays cold **or** cooler
- there is a bigger difference in temperature between the water vapour and the water
- some vapour escapes in B

*accept 'it is cooled over a bigger length **or** for longer'*
accept 'more water is collected'

1 (L6)

[7]